YOMNA BOUALLEGUE

Cloud Computing Engineering Student | Full-Stack Developer | DevOps Enthusiast

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PROFILE

2nd-year Cloud Computing Engineering student at ESPRIT, with hands-on experience in Full-Stack Development, DevOps, Kubernetes, OpenStack and Machine Learning. Built and deployed cloud-native applications using React, Spring Boot, Node.js, Docker and Azure. Seeking a software engineering, cloud or DevOps internship.

PROFESSIONAL EXPERIENCE

Intern — Full-Stack Developer | **Catalyze Tech** *July 2025 – August 2025*

FitLife — Sports & Nutrition Coaching Platform

Technologies: *React Native, React.js, Node.js, Express.js, MongoDB, JWT, Docker, Nginx, Azure VM, Figma, Postman*

- Designed and delivered a full sports and nutrition coaching application within 6 weeks, from initial concept to production deployment.
- Built a multi-role React Native mobile app (clients, coaches, nutritionists) featuring workout tracking, personalized programs, BMI/BMR calculation, geolocation and an integrated shop.
- Developed a React.js admin dashboard for centralized supervision and statistical analysis of user data.
- Designed secure REST APIs (Node.js, Express.js, JWT, MongoDB) and deployed the full stack on an Azure Linux VM using Docker and Nginx as reverse proxy.

ACADEMIC PROJECTS

TFAKADNI — Cloud-Native Maternal Health Platform *2025 – 2026*

Technologies: *OpenStack, Kubernetes, Docker, Helm, Ansible, Prometheus, Grafana, Alertmanager, Angular, Spring Boot, MySQL, JWT*

- Deployed a 7-node private OpenStack cloud on Ubuntu Server 24.04 (Keystone, Nova, Neutron/OVN, Glance, Cinder, Swift) with full automation via nested Heat Templates.
- Set up a Kubernetes v1.29.15 cluster (kubeadm, Flannel, Containerd) in a 3-tier architecture with StatefulSets, PVC, HPA and Ingress-nginx.
- Integrated a Prometheus + Grafana + Alertmanager monitoring stack and built an intelligent self-healing pipeline to automatically fix CrashLoopBackOff errors via the Kubernetes API.
- Contributed to the web platform development (Angular + Spring Boot + MySQL, secured with JWT/BCrypt) following Agile/Scrum methodology.

MedAssist AI — Medical Diagnosis Platform using Machine Learning *2025 – 2026*

Technologies: *Python, Scikit-learn, Pandas, NumPy, Flask, Gradient Boosting, XGBoost, LightGBM, Random Forest, SVM, Logistic Regression*

- Built a medical application for early detection of serious diseases: lung cancer, brain tumor, diabetes, depression and anxiety.
- Trained and benchmarked multiple ML models (Logistic Regression, Random Forest, XGBoost, LightGBM, SVM, Gradient Boosting) across 5 medical datasets using key performance metrics (accuracy, precision, recall, F1-score, AUC).
- Deployed a Flask-based prediction interface giving doctors clear, interpretable results in real time.

ChronoSerena — Distributed Medical Platform for Chronic Diseases *2024 – 2025*

Technologies: *JavaFX, Symfony, PHP, MySQL, REST API, MVC, Singleton Pattern*

- Designed a distributed platform (JavaFX desktop app + Symfony web interface) for managing patients with chronic conditions.
- Applied MVC architecture, REST APIs, Singleton design pattern and reactive forms with robust data validation.

Brush & Bargain — Mobile Art Sales and Exhibition App *2024 – 2025*

Technologies: *FlutterFlow, Dart, Firebase*

- Built a cross-platform mobile app (FlutterFlow) for buying, selling and showcasing artworks, with event management, bookings and a customer claims module.

Kadamni — Web Recruitment Platform *2023 – 2024*

Technologies: *PHP, HTML, CSS, MySQL*

- Developed a full recruitment website covering job posting, application management and administration of employer and candidate accounts.

AgroDesk — Agricultural Management Desktop App + Arduino System *2023 – 2024*

Technologies: *C++, Qt Designer, Oracle DB, Arduino, Servo Motor*

- Built an OOP C++ desktop application with Qt Designer for farm management, integrated with an Arduino module for automated animal feeding scheduling (servo motor, push button).

Banking Security System — PIC16F877 Microcontroller *2023 – 2024*

Technologies: *Embedded C, PIC16F877, EEPROM, Gas/Temperature Sensors*

- Designed an embedded banking security system on PIC16F877 monitoring gas and temperature sensors, with automatic alarm triggers and an EEPROM incident log for security agents.

Aryam — 2D Video Game (SDL / C) *2022 – 2023*

Technologies: *C, SDL*

- Developed a narrative 2D game in C/SDL where heroine Aryam fights depression during the Covid-19 lockdown, featuring a game loop, sprite rendering and collision detection.

EDUCATION

Engineering Degree — Cloud Computing (2nd year) *2023 – Present*

ESPRIT, Tunis

Scientific Baccalaureate *2021 – 2022*

TECHNICAL SKILLS

Cloud & DevOps: OpenStack, Kubernetes, Docker, Helm, Ansible, Prometheus, Grafana, Alertmanager, Azure VM, Nginx, Heat Templates, Cinder, Swift, Flannel, Ingress-nginx

Languages: C/C++, Java, Python, PHP, JavaScript, Dart/Flutter

Web & Mobile: Angular, Spring Boot, React.js, React Native, Node.js, Express.js, Flask, Symfony, JavaFX, Qt, HTML5/CSS3, FlutterFlow

Machine Learning: Scikit-learn, Pandas, NumPy, Gradient Boosting, XGBoost, LightGBM, Random Forest, SVM, Logistic Regression

Embedded Systems: Arduino, PIC16F877, Embedded C, Qt Designer, SDL

Databases: MongoDB, MySQL, Oracle, SQL

Tools: Git, GitHub, Postman, Figma, Draw.io, UML, JWT, BCrypt, Agile/Scrum

Languages: French (B2), English (B2), German (basic), Spanish (basic)